



University of Warwick Computing Society



Linux 101

But first... the Computing Society!



The Academic Society for Computer Science!

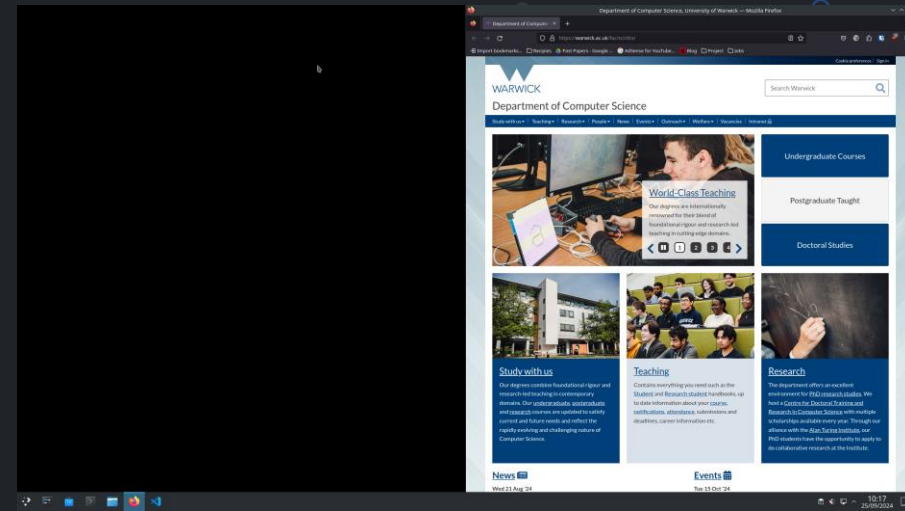
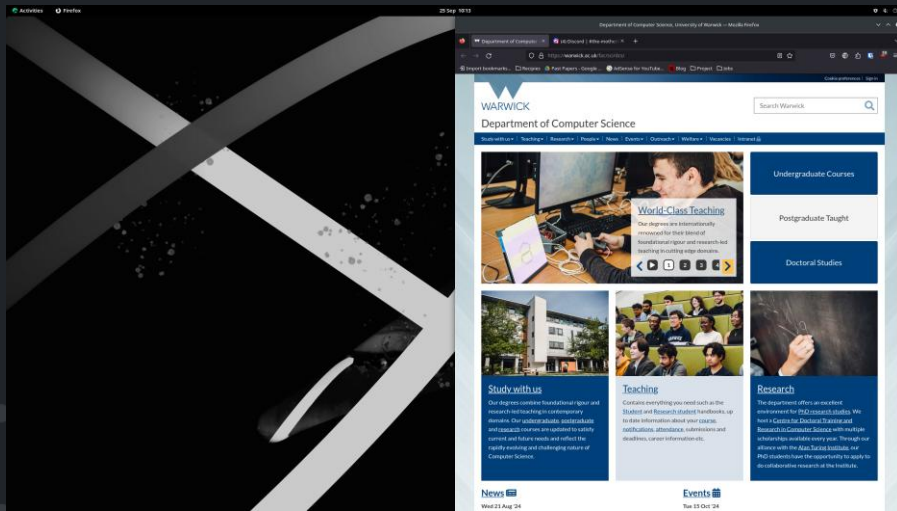
*Consider joining the Society,
checking out our Discord and
following our Instagram!*

What's Linux?

- GNU/Linux is an operating system like Windows or macOS
 - Primarily the Linux Kernel and various utility programs to interact with it
 - There are lots of different distributions, each with their own default programs and configurations
- It's distributed freely, with source code available online
 - Maintained by a community of over 22k developers
- Used by the DCS Lab computers, and most web servers.
 - Including the department's and UWCS's
 - Also popular among programmers as a desktop OS

The DCS Machines

The departmental machines run Rocky Linux 9, which should give a similar feeling to Windows or MacOS. The default desktop environment is **Gnome**



The default layout is very similar to MacOS but with the dock on the left, but you can customise it if you wish

To change between Gnome and KDE, click the settings cog on the password window

Installed Programs

These machines have familiar (and maybe unfamiliar) software installed:

Browsers



Chrome



Firefox

Text Editors



VS Code



Vim



Emacs



Kate



Atom

PDF Viewers



Okular



Evince

The Command Line

- The main way to interact with the OS
- Interact with via a terminal emulator (such as Konsole)
- A typical line in the terminal might look like:

bash-4.2\$ ls -lh ~/public_html/css

↑
Prompt

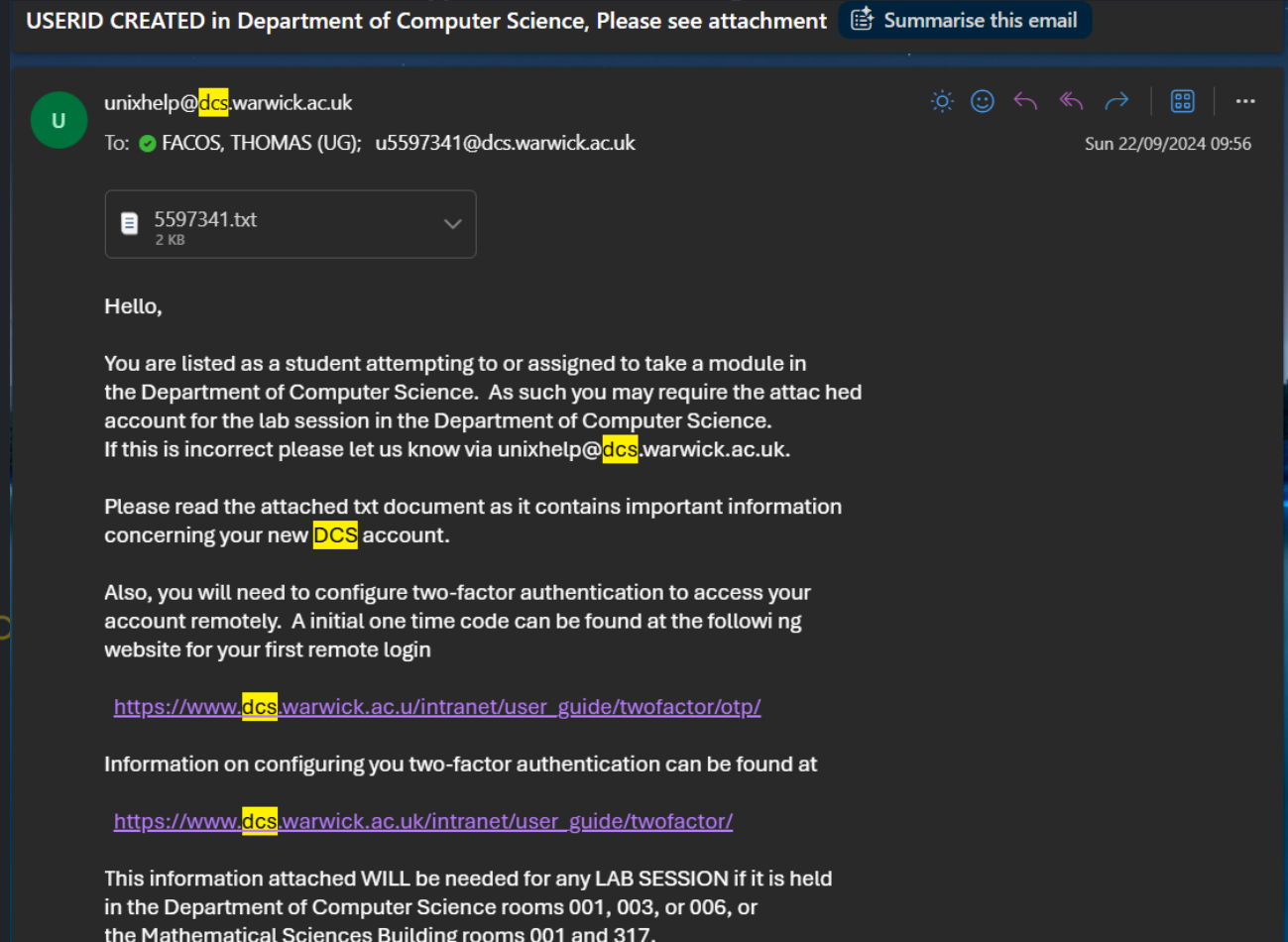
↑
Command

↑
Flags

↑
Arguments

Logging in for the First Time

- You should have already received an email from unixhelp@dcswarwick.ac.uk
- This contains instructions on how to get set-up, as well as a link to send your password for the first time



Remote Access

Sometimes, you may want to connect to the DCS Machines, but not want to have to be in the department (common reasons include laziness and/or not wanting to walk through campus at 4am)

To start set up a remote connection to the terminal, you can follow the instructions at:

<https://uwcs.co.uk/resources/remotedcs/>



We'll give a brief outline of how to connect over the next few slides.

Windows/Mac/Linux Access

- You can access DCS systems remotely using the `ssh` command, if present:

```
$ ssh uXXXXXXX@remote-YY.dcs.warwick.ac.uk
```

- Here, the `X`'s should be replaced by your university ID, and the `Y`'s should be the last two digits of it.

```
$ ssh u1234567@remote-67.dcs.warwick.ac.uk
```

- For security purposes, the password prompt will not display your input

Windows/Mac/Linux Access

- To set up 2fa (**mandatory**), run

```
$ dcs-2fa
```

- Or, set up SSH keys so you don't need to use your password
- To change your password run

```
$ passwd
```

Graphical Remote Access

- The way we described will only give you access to the command line.
- You can remotely access DCS systems with a GUI using a VNC server.
 - This will come in handy for, e.g., the CS118 coursework.
- Instructions to set this up can be found in the DCS Systems User Guide.
 - https://warwick.ac.uk/fac/sci/dcs/intranet/user_guide/remote-login/

VS Code Remote Access

- This is a middle level of access, not full desktop, but better than just terminal
- Allows using VS Code as usual as if it were on your own system, except all files and terminals are on DCS.
- Install the Remote-SSH extension and follow the instructions on <https://uwcs.co.uk/resources/remotedcs/>
 - This also includes setting up an authentication key for easier access.

The Linux Filesystem

- Linux's filesystem is a single-rooted tree
 - Removable media is mounted to this tree, not given its own tree
- Every item is considered a file, with a few variations:
 - Ordinary file, Directory (or Folder), Device file, Sockets, ...
- Directories are just files with a list of associated sub-files

Directories and Paths

- The terminal will initially load into your home directory
- To navigate around you use the **cd** command
 - The **cd** command takes a path to the directory you want to move to
- Paths in Linux are **case sensitive** and shouldn't (but may) contain spaces
- Paths can be **relative** to the current directory or **absolute**

../../foobar



Relative path

/var/www/html/



Absolute path

Directories and Paths

Linux has some special relative path symbols

•

Current Directory

• •

Parent Directory

~

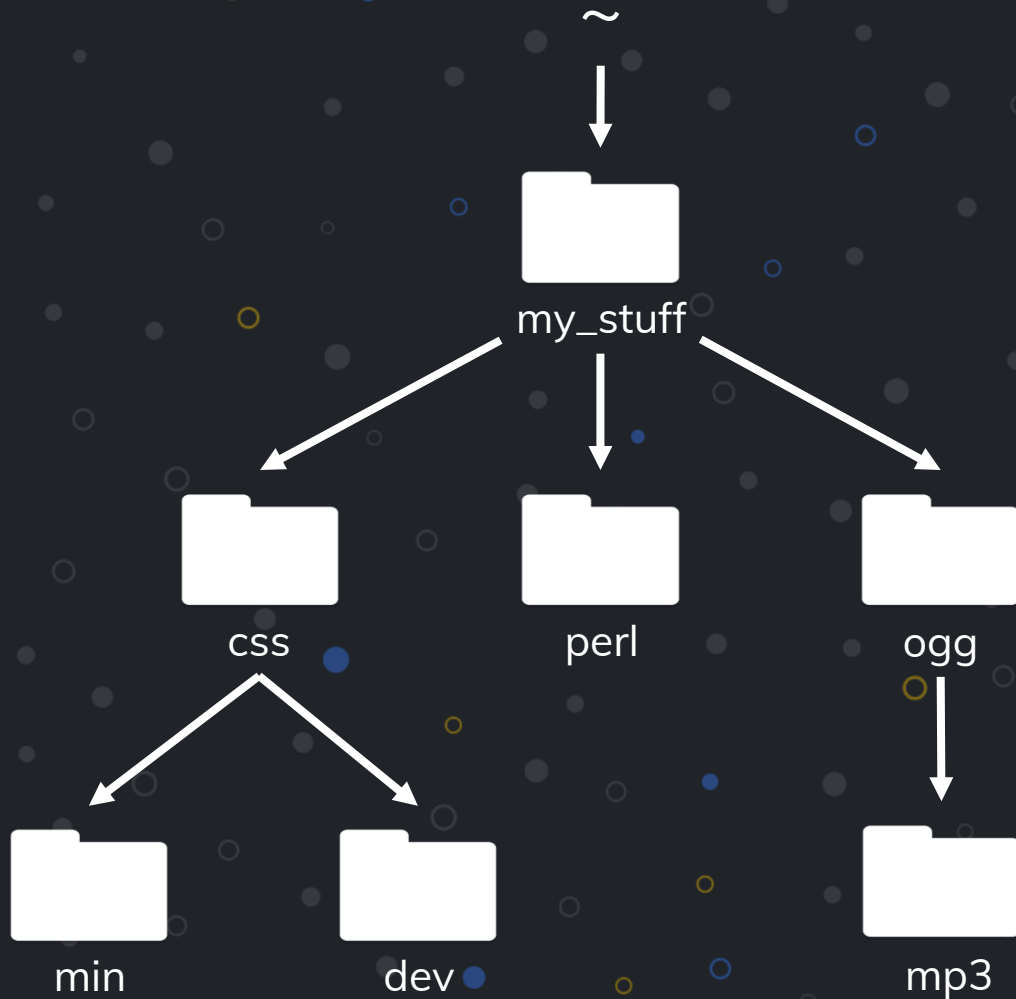
Home Directory

Example:

```
$ cd ~/foobar/./Baz/../../blat
```

```
$ cd ~/blat
```

Navigation Example



Navigation Example

~/my_stuff



Navigation Example

~/my_stuff

```
|_ css
|   |_ min
|   |_ dev
|_ perl
|_ ogg
|   |_ mp3
```

\$ cd .



Navigation Example

~/my_stuff

```
|_ css
|   |_ min
|   |_ dev
```

```
|_ perl
```

```
|_ ogg ←
```

```
    |_ mp3 ←
```

\$ cd ..

Navigation Example

~/my_stuff



```
$ cd css/min
```

No such file or directory: css/min/

Navigation Example

~/my_stuff



\$ cd ../css/min

Navigation Example

~/my_stuff



\$ cd ~/my_stuff/perl

Listing Files and Folders

- Okay, we can move around directories, but it would be nice to actually know what's in them.
- You can list files and folders using the `ls` command.
- `ls` lists the contents of the current directory by default
 - You can give `ls` a path to list that specific directory

```
$ ls ~
```

```
Desktop Documents Downloads Music Pictures  
Public README Templates Videos
```

Listing Files and Folders

To format the output into a detailed list, we can type:

```
$ ls -l
```

```
total 1337
```

```
drwxrwxrwx
```

```
3 u5XXXXXX
```

```
dc sugrad
```

```
102 31 Oct 2014 Baz
```

```
drwxr-xr-x
```

```
4 u5XXXXXX
```

```
dc sugrad
```

```
136 10 Aug 13:39 Foobar
```

↑
Permissions

↑
Owner user/group

↑
Size
(in bytes)

↑
Date modified

↑
Name

- `ls -a` includes hidden files/folders (those that begin with a dot)
- These options can be combined (`ls -al`)

Creating Files and Folders

- To make files and folders you use **touch** and **mkdir** respectively
 - Write permissions are needed in the parent folder to do this
- **touch** <file path> creates an empty file
- **mkdir** <directory path> creates an empty directory

```
$ touch foo.txt
```

```
$ mkdir -p ~/baz/blat
```

Removing Files and Folders

- `rm <path>` removes file(s) on the given path(s)
 - Write permissions are needed in the parent folder to do this
- Using `rm` is typically irreversible and does not use the 'recycle bin'
- `rm -r` recursively removes all files and folders in a given directory

```
$ rm -r ~/stuff_i_dont_need
```

- You can use the graphical 'recycle bin' in the desktop.
- Files are backed up nightly, you can get a backup restored if needed.

Manipulating Files

- You can copy a file using the **cp** command

```
$ cp original.txt ../copied_file.txt
```

- You can move a file using the **mv** command
 - This is also how you rename a file

```
$ mv original.txt ../new_dir
```

```
$ mv original.txt renamed.txt
```

Zip Archives

- Zip archives are a useful way of creating a compressed "bundle" of files and folders. You'll likely find yourself using this for coursework submissions.
- You can create a zip archive using the `zip` command

```
$ zip -r my_zip.zip ./my_unzipped_folder
```

- It is **SUPER IMPORTANT** to use `-r` when zipping – this is the recursive flag, which iterates over the full folder structure, without it, you may well find yourself creating an empty zip archive.
- Make sure to double check your submission's files before hitting the submit button!

File Permissions

- Perhaps the most confusing part of Linux
- Everything in the Linux file system is owned by a user and a group
- Users not in these categories are classed as 'other'

rwX r-X --X
↑ ↑ ↑
User Group Other

Permission	Char	Meaning
Read	r	Can view the contents
Write	w	Can change the contents
Execute	x	Can run as a program

Disk Space

- In your 1st Year, you are given 12GB of disk space by DCS
 - This increases to 15GB for 2nd year, 25GB for 3rd year, 30GB for 4th year/MSc
 - You can email the department's tech help if you require more
- You can check your remaining disk quota using **quota**
- You may not be able to log in graphically if you run out of space

```
$ quota -vs
```

```
Disk quotas for user u5XXXXXX (uid 12345):
```

Filesystem	space	quota	limit	grace	files
/export/ug_w/21	5510M	6055M	6641M		55731

Printing

- Each student in the department is given 350 credits for free printing
 - You can check your total credits by emailing tech@dcs.warwick.ac.uk
- You can print PDFs and plain text files using `lpr -P <printer> <file>`
- You can check the print queue using `lpq`
- Removing a print job is done using `lprm <job id>`
- The printers in the atrium are `dcs006a` for mono and `dcs001c` for colour

```
$ lpr -P dcs006a ~/cs132_coursework_1.pdf
```

Command Manuals & Help

- Use `man <command>` or `<command> --help` to display help menus
- These menus have in-depth information about commands
 - Optional arguments, command operation, etc
- A lot of the command information in these slides is also in man pages

```
$ man man
```

Handy Shortcuts

- Clearing the terminal window is done with `clear` or pressing `Ctrl+L`
- To copy in the terminal use `Ctrl+SHIFT+C`
- Paste with the scroll button on the mouse
- Use `pwd` to print the directory you're currently in
 - Gives the absolute file path
- To avoid typing out the full name of a file, you can use `tab`
 - Name must be unique in the directory

Help! I've F*cked Up

- Exiting a terminal program is done using **Ctrl+C** (copying can be done via Ctrl+Shift+C)
 - If that hasn't worked, try pressing 'q' or ':q' (this is to exit vim if you get stuck in it)
 - If you get this far, restart the terminal window
 - Still doesn't work? Get in touch with the DCS Systems Team

A Practical Example

We've got a lab sheet for you!

- This will put the things we've discussed today in action.
- By the end, you'll have a public webpage on the DCS website!

UWCS Linux 101 Lab

A practical introduction to Linux



Logging In

Before we start, you'll need to gain access to the department machines - the information needed for this should have been sent to your university email address. If you have any issues logging in, let a lab tutor know! They'll also be around to help if you get stuck.

Now that you've (hopefully) logged in, let's get started with the terminal!

Getting Around

First, click the colourful start button at the bottom-left corner of the screen. Then, click the "terminal" menu item. You should see something like this:

LAB TIME



Slides



Tasks



Cheatsheet

- Questions about the DCS systems? Send to tech@dcswarwick.ac.uk
- More in-depth resources are available at https://warwick.ac.uk/fac/sci/dcs/intranet/user_guide/
- Any other questions after the talk? Hop on our Discord: <https://discord.uwcs.uk> and ask questions in [#linux-101](#)